CS543 Web Services — Assignment 3

Integrate an External SOAP Partner

Project: Campus/Student Lost & Found Management System

Team ID: 5

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1. Integration Context

The Campus Lost & Found Management System integrates with an external campus verification partner to verify claims submitted by students. The partner provides a SOAP web service with the operation `verifyClaim`.

The request contains the claim reference, item description, claimant name, and optional evidence reference. The response contains the verification status, verification reference, and reason.

SOAP is used at this external integration boundary because it provides a structured XML message format and a WSDL contract describing the service interface.

2. HTTP Binding

The service uses SOAP 1.1 with document/literal style over HTTP. The client sends a SOAP request using HTTP POST.

Endpoint: https://verification.partner.example/soap/claim-verification

Content-Type: `text/xml; charset=utf-8`

SOAPAction: https://verification.partner.example/verifyClaim

The SOAP request includes partner credentials in the Header and the `VerifyClaimRequest` operation data in the Body. In a real deployment, HTTPS must be used to protect credentials and transmitted data.

3. Service Discovery

The Lost & Found development team obtains the partner's WSDL through an approved partner onboarding portal or a documented WSDL URL. The team reviews the service contract, endpoint, authentication requirements, and supported operations before integration.

The approved details are recorded in the organization's service catalogue. This assignment uses a design-time example; no live UDDI registry is required.

Illustrative registry entry:

- Business: Campus Verification Partner
- Service: CampusLostFoundVerificationService
- Operation: `verifyClaim`
- Binding: SOAP 1.1, document/literal
- Endpoint: https://verification.partner.example/soap/claim-verification
- WSDL: https://verification.partner.example/wsdl/claim-verification.wsdl

The example domain is a placeholder and does not represent a live service.

4. Fault Mapping

The integration adapter translates partner SOAP faults into errors understood by the Lost & Found application.

Partner error	Internal application response
Verification service unavailable	HTTP 503 — `CLAIM_VERIFICATION_UNAVAILABLE`
Claim could not be verified	HTTP 422 — `CLAIM_VERIFICATION_FAILED`
Invalid request	HTTP 400 — `INVALID_VERIFICATION_REQUEST`

A temporary partner outage must not be treated as a rejected claim. The application should display a safe error message and allow the user to retry later. Internal partner details and correlation identifiers should be retained only in appropriate logs.

5. Supporting Files

The submission includes:

1. `partner.wsdl` — WSDL service contract.
2. `soap-request.xml` — SOAP request containing credentials in the Header and operation data in the Body.
3. `soap-response.xml` — Successful SOAP response.
4. `soap-fault.xml` — Example SOAP fault.
5. `README.txt` — Submission file descriptions.

Conclusion

This design demonstrates how the Lost & Found system can integrate with an external verification partner using SOAP, WSDL, HTTP binding, service discovery, and fault mapping. The integration is a design-time model and does not require a running partner server.